

HE2003 Econometrics I: Linear Regression with One Regressor 3

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Linear Regression with One Regressor

Another example of selection bias: Do hospitals make people healthier?

Result of US national hospital interview survey (higher mean health status suggests better health condition):

Group	Sample Size	Mean Health Status	Std.Err.
Hospital	7,774	3.21	0.014
Non-hospital	90,049	3.93	0.003

Hospital: those who have visited hospitals during certain period given by the survey

Non-hospital: those who have not visited hospital during certain period given by the survey

Conclusion at face value: going to the hospital makes people sicker!

Reason: People who go to the hospital are probably less healthy to begin with, i.e., **Selection bias**.

Question: How to build a linear regression model for this dataset? Does the OLS estimator solve the bias problem?

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Some further examples in which **we may worry about the selection bias**:

- **Wage regression.** Wages are only observed for individuals who have wage income, which means that the individual is a member of the labor force and has a wage-paying job.
- **Program/policy evaluation.** The goal is to measure the impact of a program or a policy. **Self selection** arises when individuals volunteer to participate (rather than being randomly assigned). Individuals who volunteer for a training program may have abilities/features correlated with outcomes.
- **Surveys.** While a survey may be randomly distributed the act of completing the survey is non-random. Most survey have response rates considerably below 100%.
- **Ratings.** We are routinely asked to rate products, services, and experiences. Most people do not respond to the request.

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More Explanations on the Least Square Assumption 2:

$(X_i, Y_i), i = 1, \dots, n$ are independently and identically distributed (i.i.d.).

- In statistics and econometrics, it is important to distinguish the **difference between population and sample**.
- For instance, suppose that you are interested in the **distribution of wages in Singapore** (e.g., average wage, wage difference (spread) among workers). It may be **difficult to interview all the workers** to obtain the whole distribution.
- Thus, we typically **estimate** on the basis of a group of **randomly selected people** (say, 1000, 5000, etc). In this example, the wages of all workers in Singapore is the **population**, while the wages of the 1000 randomly selected people is the **sample**.

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Here, a **key idea** is that we view the wage of a worker as a **random variable**. This **does not mean** that his/her **wage fluctuates randomly**. What is meant is that the **selection procedure is random**: He/she might or might not be selected into the sample. **Had a different sample be drawn**, then the observations would have been different.

Population (left) vs Sample (right)



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KEY CONCEPT: The Mechanism of Simple Random Sampling

- Suppose that n observations $Y_1, \dots, Y_n$ are **selected randomly from a population** of interest (i.e., n samples from the population).
- Under a **simple random sampling**, each member of the population is **equally likely to be drawn**. Thus, knowing the value of Y_1 provides no information about Y_2 (or $Y_3, Y_4, \dots$), so $Y_1, \dots, Y_n$ are **independent from each other**.
- Because $Y_1, \dots, Y_n$ are randomly drawn from the **same population** (e.g., all workers in Singapore), the **distribution of Y_i is the same** for each $i = 1, \dots, n$.

Therefore, $Y_1, \dots, Y_n$ are **independently and identically distributed** (i.i.d.).

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The importance of random sampling

Suppose that, to estimate the **monthly national unemployment rate**, we adopt a sampling scheme in which interviewers **survey working-age adults sitting in city parks at 10am on Wednesdays** of the month. Then, the **unemployed would be overly represented** in the sample, and an estimate of the unemployment rate based on this sampling plan would be **biased**.

Example (The U.S. current population survey (CPS))

- Each month, the U.S. Census Bureau and the U.S. Bureau of Labor Statistics conduct the CPS, which provides data on **labor force characteristics of the U.S. population**, including the **levels of employment, unemployment, and earnings**.
- Approximately **6,000 U.S. households** are surveyed each month. The sample is chosen by **randomly selecting addresses** from a database of addresses. In particular, the CPS **earnings data** are for **full-time workers**, defined to be somebody employed more than 35 hours per week for at least 48 weeks in the previous year.

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The main place we will encounter **non-i.i.d. sampling** is when data are **recorded over time** (e.g., **time series data**).

For example, we might have data on the **revenue/profit** (Y) of a firm and the **interest rate** at which the firm can borrow (X), where these data are collected over time from a specific firm; e.g., they might be recorded four times a year (quarterly) for 30 years. This is an example of **time series data**, which tend to be **correlated with each other (auto-correlated)**.

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Why are the 3 OLS Assumptions important?

Under the 3 Assumptions, the OLS estimator has **good properties**:

- $E[\hat{\beta}_1]$ and $E[\hat{\beta}_0]$? (measure of **center**)
 $E[\hat{\beta}_1] = \beta_1$ and $E[\hat{\beta}_0] = \beta_0$: the OLS is **unbiased**. Thus, the distributions of $\hat{\beta}_1$ and $\hat{\beta}_0$ are **centered at β_1 and β_0** , respectively.
- $Var[\hat{\beta}_1]$ and $Var[\hat{\beta}_0]$? (measure of **spread**)
 $Var[\hat{\beta}_1]$ and $Var[\hat{\beta}_0]$ are **relatively small** compared with the variances of alternative estimators. That is, $\hat{\beta}_1$ and $\hat{\beta}_0$ are **more efficient** than alternative estimators (If an estimator has **smaller variance**, then it **uses the information** in the data **more efficiently**).
- What is the distribution of $\hat{\beta}_1$ and $\hat{\beta}_0$ in **large sample**?
($n = 100, 500, 1000, \dots$) They are **normally distributed**.

In sum, the distribution of the OLS estimator is **centered at the true value**, with relatively **small spread**. And it has a well-behaved **bell curve in large sample**. We will provide more details later.

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KEY CONCEPT: Measure of Fit, R^2 .

- The **coefficient of determination** or R^2 is often used as a measure of fit.
- The R^2 measures the **fraction of the sample variation in Y that is explained by X** . Notice that Y_i can be decomposed as

$$Y_i = \hat{Y}_i + \hat{u}_i,$$

where $\hat{Y}_i$ is the **OLS predicted value** and $\hat{u}_i$ is the **residual**. R^2 is the ratio of the sample variance of $\hat{Y}$ to the sample variance of Y .

- Specifically, the R^2 can be written as the ratio of the **explained sum of squares (ESS)** to the **total sum of squares (TSS)**:

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2,$$

$$TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

Notice that $n^{-1} \sum_{i=1}^n \hat{Y}_i = n^{-1} \sum_{i=1}^n Y_i$ because $\sum_{i=1}^n \hat{u}_i = 0$.

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Then, the R^2 is defined as:

$$R^2 = \frac{ESS}{TSS}.$$

- The R^2 ranges between 0 and 1.
- If $\hat{\beta}_1 = 0$, then X explains none of the variation of Y and the predicted value of Y_i is $\hat{Y}_i = \hat{\beta}_0 = \bar{Y}$. In this case, the ESS is zero and the R^2 is zero.
- In contrast, if X explains all of the variation of Y , then $Y_i = \hat{Y}_i$ for all i and every residual is zero, so that $ESS = TSS$ and $R^2 = 1$.
- In general, the R^2 does not take on the extreme value of 0 or 1 but falls somewhere in between. An R^2 near 1 indicates that X is good at predicting Y , while an R^2 near 0 indicates that X is not very good at predicting Y .

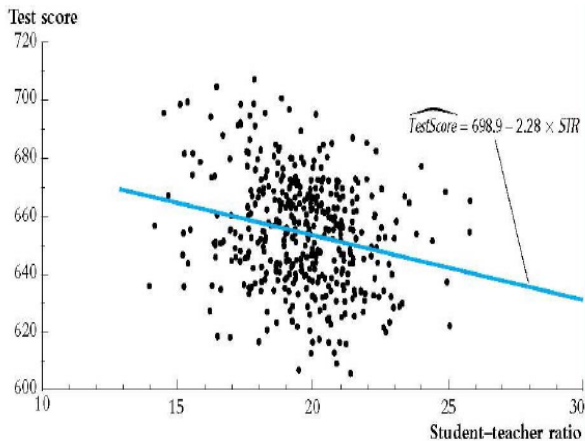
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Application to the Test Score Data

- The R^2 of the OLS regression is 0.051, or 5.1%.
- 5.1% means that the explanatory variable *STR* explains 5.1% of the variance of the dependent variable *TestScore*. As the scatterplot shows, the student-teacher ratio (STR) explains some of the variation in test scores, but much variation remains unaccounted for.
- What should we make of the low R^2 ? This does not imply that this regression is “good” or “bad”. What the low R^2 does tell us is that other important factors influence test scores.
- These factors could include differences in the student body across districts, differences in school quality unrelated to the STR, or luck on the test, etc.
- The low R^2 does not tell us what these factors are, but it does indicate that the STR alone explains only a small part of the variation in test scores.

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$R^2 = 0.051$: The STR explains some of the variation in test scores, but much variation remains unaccounted for.



Linear Regression with One Regressor

KEY CONCEPT: Distribution of the OLS Estimators in Large Samples

- Because the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are computed from a randomly drawn sample, **the estimators themselves are random variables with a probability distribution**, which describes the values $\hat{\beta}_0$ and $\hat{\beta}_1$ could take over different possible random samples.
- This is similar to the sample average $\bar{Y}$, the estimator of the population mean μ . **Because $\bar{Y}$ is calculated using a randomly drawn sample, $\bar{Y}$ is a random variable that takes on different values from one sample to the next** (as we have seen in the computer simulation); the probability of these different values is summarized in the distribution of $\bar{Y}$.
- If the OLS assumptions hold, then **in large samples $\hat{\beta}_0$ and $\hat{\beta}_1$ follow approximately normal distributions** (by Central Limit Theorem). Typically, $n > 100$ is sufficiently large for the distributions of the OLS estimators to be well approximated by normal distributions.

Linear Regression with One Regressor

The details of the unbiasedness and efficiency are complicated for the OLS because of the formula of $\hat{\beta}_1$ and $\hat{\beta}_0$. **For simplicity**, we look at the **sample average** as an example. Note that the sample average is an estimator of the population mean.

The **sample average** $\bar{Y}$ of $Y_1, \dots, Y_n$ is defined as:

$$\bar{Y} = \frac{1}{n} (Y_1 + \dots + Y_n) = \frac{1}{n} \sum_{i=1}^n Y_i.$$

Because $Y_1, \dots, Y_n$ are random variables, the **sample average is also a random variable**. Thus, it also has **expected value** and **variance**.

KEY CONCEPT: The expected value and variance of $\bar{Y}$

Let $Y_1, \dots, Y_n$ be i.i.d. with $E[Y_i] = \mu$ and $\text{Var}[Y_i] = \sigma^2$.

Then, the expected value of $\bar{Y}$ is:

$$E[\bar{Y}] = E\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n} E\left[\sum_{i=1}^n Y_i\right] = \frac{1}{n} \sum_{i=1}^n E[Y_i] = \mu.$$

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The variance of $\bar{Y}$ is:

$$\begin{aligned}\text{Var}[\bar{Y}] &= \text{Var}\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] \\&= \frac{1}{n^2} \text{Var}\left[\sum_{i=1}^n Y_i\right] \\&= \frac{1}{n^2} [\text{Var}(Y_1) + \text{Var}(Y_2) + \dots + \text{Var}(Y_n)] \\&= \frac{1}{n^2} [\sigma^2 + \sigma^2 + \dots + \sigma^2] \\&= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n},\end{aligned}$$

where the **third equality** follows from the fact that all the **covariance terms** (such as $\text{Cov}(Y_1, Y_2)$, $\text{Cov}(Y_1, Y_3)$, $\text{Cov}(Y_2, Y_3)$, ...) are **equal to zero** because $Y_1, \dots, Y_n$ are **independent from each other**.

Linear Regression with One Regressor

- Given i.i.d. observations $Y_1, \dots, Y_n$, the most natural estimator for the population mean μ would be $\bar{Y}$, the **sample mean** (or sample average).
- However, $\bar{Y}$ **is not the unique estimator** of μ . For instance, Y_1 (the first observation) or $(Y_1 + Y_n)/2$ (the average of the first and last observations) can also be an estimator of μ .

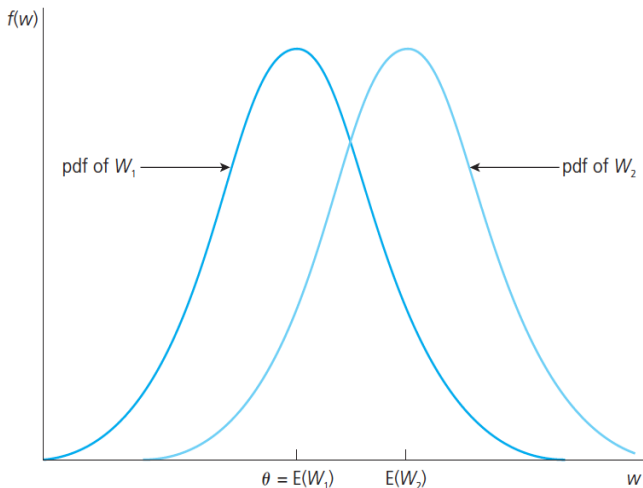
Then, **why shall we use $\bar{Y}$?**

KEY CONCEPT: Unbiasedness and efficiency

Let $\tilde{Y}$ be an estimator of the population mean μ . The estimator is “good” if it satisfies the following properties:

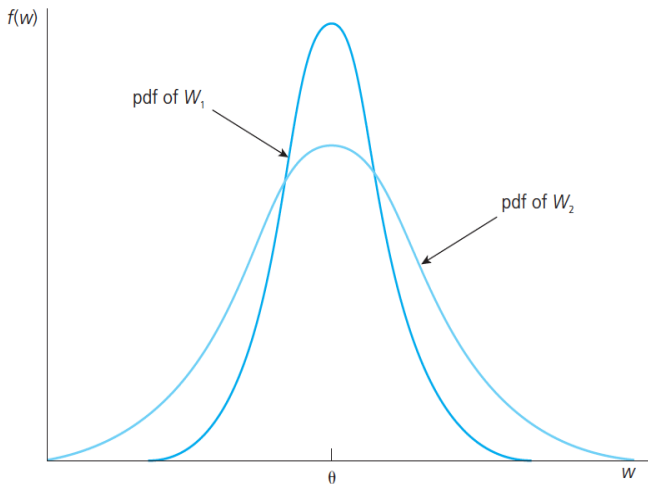
- **Unbiasedness:** $E[\tilde{Y}] = \mu$.
- **Efficiency:** Let $\hat{Y}$ be an alternative unbiased estimator for μ . If $\text{Var}[\tilde{Y}] < \text{Var}[\hat{Y}]$, then $\tilde{Y}$ is said to be **more efficient** than $\hat{Y}$.

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The distributions of an unbiased estimator ($E[W_1] = \theta$) and biased estimator ($E[W_2] \neq \theta$); The parameter of interest is θ .

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The distributions of two unbiased estimators ($E[W_1] = \theta$, $E[W_2] = \theta$); W_1 has smaller variance (is more efficient) than W_2 .